

系統程式設計

Lab5 Stdio Compare

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1. Where is the counter you add?
(請註明在程式碼的第幾行，最好加上截圖)

On line 7 & line 10

```
1  #include "apue.h"
2
3  #define ll long long
4
5  int main(void) {
6      int c;
7      ll counter = 0;
8
9      while ((c = getc(stdin)) != EOF) {
10         counter++;
11
12         if (putc(c, stdout) == EOF)
13             err_sys("output error");
14     }
15
16     if (ferror(stdin))
17         err_sys("input error");
18
19     fprintf(stderr, "Loop iterations (getc/putc): %lld", counter);
20
21     exit(0);
22 }
```

On line 7 & line 27

```
1  #include "apue.h"
2
3  #define ll long long
4
5  int main(void) {
6      char buf[MAXLINE];
7      ll counter = 0;
8
9      #ifndef BUF_MODE
10     #define BUF_MODE _IONBF
11 #endif
12
13     if (setvbuf(stdin, NULL, BUF_MODE, MAXLINE) != 0)
14         err_sys("setvbuf std error");
15     if (setvbuf(stdout, NULL, BUF_MODE, MAXLINE) != 0)
16         err_sys("setvbuf stdout error");
17
18     #if BUF_MODE == _IOFBF
19         fprintf(stderr, "Fully Buffered\n");
20     #elif BUF_MODE == _IOLBF
21         fprintf(stderr, "Line Buffered\n");
22     #elif BUF_MODE == _IONBF
23         fprintf(stderr, "Unbuffered\n");
24     #endif
25
26     while (fgets(buf, MAXLINE, stdin) != NULL) {
27         counter++;
28         if (fputs(buf, stdout) == EOF)
29             err_sys("output error");
30     }
31     if (ferror(stdin))
32         err_sys("input error");
33
34     fprintf(stderr, "Loop iterations: %lld\n", counter);
35
36     exit(0);
37 }
```

2. Screen shot for Table1

(提供每個執行後的截圖，並說明每張截圖是屬於哪種function)

Terminal for time on getc/putc

```
lk@LK Lab5 % make -f makefile_5_4
gcc -c -o fig5_4.o fig5_4.c
gcc -c -o error.o error.c
gcc -o exe_fig5_4 fig5_4.o error.o
lk@LK Lab5 % time ./exe_fig5_4 < 100mb_file > /dev/null
Loop iterations (getc/putc): 104857600./exe_fig5_4 < 100mb_file > /dev/null  5.85s user 0.12s system 88% cpu 6.783 total
```

Terminal for time on fgets/puts

```
lk@LK Lab5 % make -f makefile_5_5
gcc -c -o fig5_5.o fig5_5.c
gcc -c -o error.o error.c
gcc -o exe_fig5_5 fig5_5.o error.o
lk@LK Lab5 % time ./exe_fig5_5 < 100mb_file > /dev/null
Loop iterations (fgets/fputs): 25607./exe_fig5_5 < 100mb_file > /dev/null  0.01s user 0.05s system 11% cpu 0.557 total
```

3. Screen shot for Table2

(提供每個執行後的截圖，並說明每張截圖是屬於哪種mode)

Fully buffered mode

```
lk@LK Lab5 % time ./exe_fig5_5 < 100mb_file > /dev/null
Fully Buffered
Loop iterations: 25607
./exe_fig5_5 < 100mb_file > /dev/null 0.01s user 0.05s system 11% cpu 0.553 total
```

Line buffered mode

```
lk@LK Lab5 % time ./exe_fig5_5 < 100mb_file > /dev/null
Line Buffered
Loop iterations: 25607
./exe_fig5_5 < 100mb_file > /dev/null 0.02s user 0.05s system 11% cpu 0.528 total
```

Unbuffered mode

```
lk@LK Lab5 % time ./exe_fig5_5 < 100mb_file > /dev/null
Unbuffered
Loop iterations: 25607
./exe_fig5_5 < 100mb_file > /dev/null 23.32s user 63.25s system 98% cpu 1:28.21 total
```